The Charged Lepton Mass Spectrum from 600-Cell Lattice Geometry

Conscious Point Physics — SM-6 (Version 1)

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Abstract

The charged lepton mass spectrum is derived from the geometry of the 600-cell polytope with one calibration constant (the overall mass scale $\mathrm{SSV}_0 = m_e c^2/2$) and zero free shape parameters. The derivation has two independently verifiable parts.

Part I: The Weinberg angle is derived from the spectral traces of the 600-cell adjacency matrix A. The bare topological mixing ratio $\mathrm{Tr}(A^2)/[\mathrm{Tr}(A^2) + \mathrm{Tr}(A^3)/3] = 3/8$ is proved from mode counting: $\mathrm{Tr}(A^2) = 2E = 1440$ counts abelian edge modes ($\mathrm{U}(1)_Y$) and $\mathrm{Tr}(A^3)/3 = 2F = 2400$ counts non-abelian face-circulation modes ($\mathrm{SU}(2)_L$). The ratio 3/8 is unique to the 600-cell among all six regular 4-polytopes and equals the SU(5) GUT-scale Weinberg angle. The physical value is obtained by multiplying by the edge propagation efficiency $\eta = l_{\mathrm{edge}}/R_{\mathrm{circ}} = 1/\varphi$ (SSV/PSR metric correction), giving $\sin^2\theta_W = 3/(8\varphi) \approx 0.2318$ (PDG: 0.23121, agreement 0.24%).

Part II: The Koide phase is derived from the isotropic electroweak shift on the K_3 cage face. The base value $\cos\theta_0 = -K = -2/3$ is the K_3 eigenvalue ratio (proved in SM-3). The electroweak correction $\varepsilon = 2\sin^2\theta_W/(z+1) = 3/(52\varphi)$ shifts all K_3 eigenvalues isotropically, changing the eigenvalue ratio without breaking C_3 symmetry. This gives $\cos\theta_{\mathrm{Koide}} = -(2/3)(1 + \sin^2\theta_W/(z+1)) = -(2/3)(1 + 3/(104\varphi))$, yielding $\theta = 132.731$ (PDG: 132.732, agreement 0.003%).

Combined with the Koide parametrisation and electron mass calibration, the predicted muon mass is 105.47 MeV (PDG: 105.66, 0.18%) and the predicted tau mass is 1774.1 MeV (PDG: 1776.9, 0.15%). The Standard Model requires three independent lepton masses. The Koide formula requires two parameters (K and θ). This derivation requires one calibration constant.

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1 Introduction

The masses of the three charged leptons — electron ($m_e = 0.511$ MeV), muon ($m_\mu = 105.66$ MeV), and tau ($m_\tau = 1776.9$ MeV) — are free parameters of the Standard Model. In 1981, Yoshio Koide [1] noticed that the combination

$$K \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{1}$$

is satisfied to 11 ppm by the measured masses. This relation has remained unexplained for 44 years. Its two parameters — the ratio $K = 2/3$ and the Koide phase $\theta = 132.73$ that determines the individual masses — have no known origin in the Standard Model.

In Conscious Point Physics (CPP) [2–4], the ratio $K = 2/3$ was derived from the eigenvalue spectrum of the K_3 complete graph (the triangular base of the tetrahedral lepton cage) [3]. However, the Koide phase θ remained a calibrated parameter.

This paper derives both K *and* θ from the 600-cell lattice geometry with zero free shape parameters, and additionally derives the Weinberg angle $\sin^2\theta_W$ as an intermediate result. The derivation uses only the six CPP axioms and the combinatorial/metric properties of the 600-cell.

2 The 600-Cell Lattice

The 600-cell is the regular 4-polytope with 120 vertices ($V = 120$), 720 edges ($E = 720$), 1200 triangular faces ($F = 1200$), and 600 tetrahedral cells ($C = 600$). Every vertex has coordination number $z = 12$. The adjacency matrix A is the 120×120 symmetric matrix with $A_{ij} = 1$ when vertices i and j are connected by an edge, and $A_{ij} = 0$ otherwise.

The edge length in circumradius units is

$$l_{\text{edge}} = \frac{1}{\varphi}, \qquad \varphi = \frac{1+\sqrt{5}}{2} \approx 1.6180 \quad \text{(golden ratio)}. \tag{2}$$

2.1 Corrected eigenvalue spectrum

The adjacency matrix has nine distinct eigenvalues (not six, as stated in earlier CPP publications). The multiplicities equal $(\dim\rho)^2$ for the nine irreducible representations of the binary icosahedral group $2I$ (order 120):

Table 1: Eigenvalues of the 600-cell adjacency matrix, verified by explicit numerical diagonalisation.

λ	Exact form	Multiplicity	$\dim\rho$
12.000	12	1	1
9.708	6φ	4	2
6.472	4φ	9	3
3.000	3	16	4
0.000	0	25	5
−2.000	−2	36	6
−2.472	$-4\varphi^{-1}$	9	3
−3.000	−3	16	4
−3.708	$-6\varphi^{-1}$	4	2

Key spectral identities:

$$\mathrm{Tr}(A) = 0, \tag{3}$$
$$\mathrm{Tr}(A^2) = 2E = 1440, \tag{4}$$
$$\mathrm{Tr}(A^3) = 6F = 7200. \tag{5}$$

Identity (4) counts closed walks of length 2 (back-and-forth along edges). Identity (5) counts closed walks of length 3 (oriented triangular face circulations).

3 Part I: The Weinberg Angle

3.1 Mode counting: the bare ratio 3/8

Definition 3.1 (Edge and face modes)**.** *An* edge mode *is a closed walk of length 2 on the 600-cell graph (a DI-bit hops along an edge and returns). This is the abelian U(1)$_Y$ propagation channel. A* face mode *is a closed walk of length 3 (a DI-bit circulates around a triangular face with consistent handedness). This is the non-abelian SU(2)$_L$ propagation channel.*

Theorem 3.2 (Bare Weinberg angle from spectral traces)**.** *The bare electroweak mixing ratio on the 600-cell is*

$$\frac{\mathrm{Tr}(A^2)}{\mathrm{Tr}(A^2) + \mathrm{Tr}(A^3)/3} = \frac{1440}{1440 + 2400} = \frac{1440}{3840} = \frac{3}{8}. \tag{6}$$

Proof. From the spectral identities (4) and (5). □ □

Remark 3.3 (Uniqueness)**.** *The ratio $E/(E+F) = 3/8$ is unique to the 600-cell among all six regular 4-polytopes (Table 2). The dual 120-cell gives $5/8 = 1 - 3/8$. In the SU(5) Grand Unified Theory, the Weinberg angle at the unification scale is $\sin^2\theta_W = 3/8$.*

Table 2: Edge-to-face ratio $E/(E+F)$ for all regular 4-polytopes.

Polytope	E	F	$E/(E+F)$
5-cell	10	10	1/2
8-cell	32	24	4/7
16-cell	24	32	3/7
24-cell	96	96	1/2
120-cell	1200	720	5/8
600-cell	**720**	**1200**	**3/8**

3.2 The propagation efficiency correction

The bare ratio 3/8 counts modes without regard to geometry. The physical mixing requires a correction for the fact that edge modes and face modes traverse different physical distances per hop.

Definition 3.4 (Propagation efficiency)**.** *The* propagation efficiency *of a mode is the physical distance covered per hop, normalised by the circumradius:*

$$\eta = \frac{l_{\mathrm{edge}}}{R_{\mathrm{circ}}} = \frac{1}{\varphi}. \tag{7}$$

Edge modes (single-hop, abelian) have efficiency $\eta = 1/\varphi$. Face modes (triangular circulation, non-abelian) operate at the circumradius scale and have efficiency normalised to 1.

Definition 3.5 (Operational Weinberg angle in CPP)**.** *The Weinberg angle is the fraction of vacuum disturbances that propagate as photons (edge-mode excitations), with each edge channel weighted by its propagation efficiency:*

$$\sin^2\theta_W \equiv \frac{\eta \cdot \mathrm{Tr}(A^2)}{\mathrm{Tr}(A^2) + \mathrm{Tr}(A^3)/3}. \tag{8}$$

The denominator is the total combinatorial mode capacity of the vacuum lattice — a topological invariant that does not depend on the metric. The numerator is the metric-corrected edge-mode throughput.

Theorem 3.6 (Weinberg angle from the 600-cell)**.**

$$\sin^2\theta_W = \frac{3}{8\varphi} \approx 0.23176. \tag{9}$$

Proof. Substitute (7) and the spectral identities (4),(5) into (8):

$$\sin^2\theta_W = \frac{(1/\varphi) \times 1440}{3840} = \frac{1}{\varphi} \times \frac{3}{8} = \frac{3}{8\varphi}. \tag{10}$$

□ □

Remark 3.7 (Comparison with PDG)**.** *PDG 2026:* $\sin^2\theta_W(M_Z) = 0.23121 \pm 0.00004$*. Agreement:* 0.24%*. The* 0.24% *residual is predicted to arise from finite-temperature Dipole Sea fluctuations (partner-switching jitter, rogue-wave SSV spikes). The lattice value* $3/(8\varphi)$ *is the zero-temperature prediction.*

Remark 3.8 (Why the denominator is fixed)**.** *In the Standard Model, the Weinberg angle is defined through gauge couplings:* $\sin^2\theta_W = g'^2/(g^2 + g'^2)$*, where both couplings are dynamic. In CPP, the mixing is between* graph modes *(topological objects) with* metric-weighted efficiencies *(geometric corrections). The mode counts* $\mathrm{Tr}(A^2)$ *and* $\mathrm{Tr}(A^3)$ *are topological invariants of the graph; the metric affects only the* efficiency *of each mode, not the* number *of modes. This is why the denominator is the fixed lattice capacity 3840, and why the* $1/\varphi$ *factor enters as a linear prefactor on the mode fraction rather than quadratically through a coupling ratio.*

4 Part II: The Koide Phase

4.1 The K_3 eigenvalue structure and the Koide ratio

The tetrahedral cage of a charged lepton has a triangular base graph K_3 (the complete graph on three vertices). Its adjacency matrix is

$$H_0 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad \lambda_+ = 2 \text{ (bonding)}, \quad \lambda_- = -1 \text{ (antibonding, double)}. \tag{11}$$

The Koide ratio follows from the eigenvalue ratio [3]:

$$K = \frac{\lambda_+}{\lambda_+ + |\lambda_-|} = \frac{2}{2+1} = \frac{2}{3}. \tag{12}$$

The Koide phase θ parametrises the individual lepton masses via [1] $\sqrt{m_i} = (S/3)(1+\sqrt{2}\cos(\theta + 2\pi i/3))$, where $S = \sum_i \sqrt{m_i}$.

4.2 The base value: $\cos\theta_0 = -K$

Proposition 4.1 (Unified origin of K and θ)**.** *The Koide ratio K and the Koide phase θ share a common origin in the K_3 eigenvalue ratio:*

$$\cos\theta_0 = -K = -\frac{2}{3}. \tag{13}$$

This gives $\theta_0 = \arccos(-2/3) = 131.81$, which is 0.92 from the PDG-derived value $\theta = 132.73$. The small correction comes from the electroweak sector.

4.3 The isotropic electroweak shift

Theorem 4.2 (Self-energy isotropy on K_3 faces)**.** *The self-energy $\Sigma(\omega) = A_{fr}(\omega I - A_{rr})^{-1}A_{rf}$ of the 600-cell, restricted to any K_3 face, is exactly proportional to the identity in the antibonding subspace at all non-resonant frequencies. No direction in the antibonding subspace is selected by the graph structure.*

Proof. Verified by explicit computation of the Green's function for four non-equivalent faces of the 1200-face 600-cell. Off-diagonal elements in the antibonding block are zero to machine precision ($< 10^{-14}$). □ □

Theorem 4.3 (Koide phase from isotropic EW shift)**.** *The electroweak sector produces an isotropic perturbation on each K_3 face:*

$$\delta H = \varepsilon\, I_3, \qquad \varepsilon = \frac{2\sin^2\theta_W}{z+1} = \frac{3}{52\varphi}, \tag{14}$$

where $z + 1 = 13$ is the closed-neighbourhood size. The Koide phase is

$$\cos\theta_{\text{Koide}} = -\frac{2}{3}\left(1 + \frac{\sin^2\theta_W}{z+1}\right) = -\frac{2}{3}\left(1 + \frac{3}{104\varphi}\right). \tag{15}$$

Proof. The proof has two parts: a bond-counting derivation of ε and an algebraic chain from ε to θ.

Bond-counting derivation of ε.

Step A. The electroweak abelian fraction at each lattice site is $\sin^2\theta_W$ (Theorem 3.6). Distributed isotropically over all $z + 1 = 13$ bonds in the closed neighbourhood: abelian energy per bond $= \sin^2\theta_W$.

Step B. Each K_3 vertex has 2 internal bonds (edges to the other two face vertices) and 10 external bonds. The EW correction assigned to K_3 per vertex is $\delta e = 2\sin^2\theta_W$.

Step C. Normalise by the total coupling per vertex $z + 1 = 13$ (the diagonal of the closed-neighbourhood Laplacian $\tilde{L} = (z{+}1)I - \tilde{A}$): $\varepsilon = 2\sin^2\theta_W/(z{+}1)$.

Step D. By Theorem 4.2, the correction is the same at all three K_3 vertices: $\delta H = \varepsilon\, I_3$.

Algebraic chain from ε to θ.

Step 1. Perturbed eigenvalues: $\lambda'_+ = 2 + \varepsilon$, $\lambda'_- = -1 + \varepsilon$, $|\lambda'_-| = 1 - \varepsilon$.

Step 2. Perturbed Koide ratio: $K' = (2{+}\varepsilon)/((2{+}\varepsilon) + (1{-}\varepsilon)) = (2{+}\varepsilon)/3$.

Step 3. Koide phase: $\cos\theta = -K' = -(2{+}\varepsilon)/3$.

Step 4. Expand: $-(2+\varepsilon)/3 = -(2/3)(1 + \varepsilon/2) = -(2/3)(1 + \sin^2\theta_W/(z+1))$.

Step 5. Substitute $\sin^2\theta_W = 3/(8\varphi)$: $\cos\theta = -(2/3)(1 + 3/(104\varphi))$. □ □

Remark 4.4 (Why this preserves C_3)**.** *The isotropic shift εI_3 does not break C_3 symmetry — eigenvectors are unchanged and the antibonding degeneracy is preserved. The Koide phase shifts because the eigenvalue* ratio $K = \lambda_+/(\lambda_+ + |\lambda_-|)$ *responds nonlinearly to a uniform shift: λ_+ increases while $|\lambda_-|$ decreases. This is consistent with Theorem 4.2: no direction is selected in the antibonding subspace, yet the Koide phase is non-trivial.*

5 Predicted Lepton Masses

Using the Koide parametrisation with the derived θ:

$$\sqrt{m_i} = \frac{S}{3}\left(1 + \sqrt{2}\cos\left(\theta + \frac{2\pi i}{3}\right)\right), \qquad \theta = 132.731, \tag{16}$$

where $S = \sum_i \sqrt{m_i}$ is fixed by the single calibration $\mathrm{SSV}_0 = m_e c^2/2 = 0.2555$ MeV (equivalent to calibrating m_e):

Table 3: Predicted charged lepton masses from the 600-cell lattice.

Lepton	Predicted (MeV)	PDG 2026 (MeV)	Agreement
Electron	0.511	0.511	calibrated
Muon	105.47	105.66	0.18%
Tau	1774.1	1776.9	0.15%

$K = 2/3$ exactly (by construction from the Koide parametrisation).

Table 4: Parameter count comparison.

Framework	Free parameters
Standard Model	3 (m_e, m_μ, m_τ independent)
Koide formula (1981)	2 (K unexplained, θ calibrated)
This work	**1** (SSV_0 only; K and θ derived)

6 Mutual Reinforcement

The Weinberg angle can be derived independently from the Koide phase by inverting (15):

$$\sin^2\theta_W = (z+1)\left(\frac{\cos\theta_{\mathrm{PDG}}}{-K} - 1\right) = 0.2322. \tag{17}$$

This agrees with the 600-cell spectral value $3/(8\varphi) = 0.2318$ to 0.19%. Two completely independent physical quantities — lepton masses and electroweak mixing — point to the same geometric constant on the 600-cell lattice.

The lattice-derived $\sin^2\theta_W = 3/(8\varphi)$ gives *better* lepton mass predictions than the PDG-measured $\sin^2\theta_W = 0.23121$ (muon: 0.18% vs. 0.39%), suggesting that the lattice value is the zero-temperature “crystalline” Weinberg angle and the PDG value includes finite-temperature Dipole Sea corrections.

7 Assumptions and Attack Surface

For clarity, we list every assumption in the derivation and its status:

1. **The 600-cell lattice** (AXIM-2). This is the core geometric hypothesis of CPP. The entire derivation follows from the combinatorial and metric properties of this polytope. If the lattice is wrong, the derivation fails.

2. **Spectral trace identities** (Eqs. 4, 5). These are standard results in algebraic graph theory. They are *proved*, not assumed.

3. **Edge modes = abelian, face modes = non-abelian** (Definition 1). This identification is the physical content of Part I. It is motivated by the 1D (linear, abelian) vs. 2D (circulatory, non-abelian) structure of the walks and by the K_3 face circulation generating SU(2). It is *physically motivated* but could in principle be challenged.

4. **Propagation efficiency** $\eta = l/R = 1/\varphi$ (Eq. 7). This uses the SSV/PSR metric from SR-1. The ratio $l/R = 1/\varphi$ is a *geometric fact* of the 600-cell. The assumption is that the efficiency is proportional to the physical distance per hop, which is the standard DI-bit hopping amplitude from QM-1.

5. **Operational definition of** $\sin^2\theta_W$ (Definition 3.5). CPP defines the Weinberg angle as (edge throughput)/(total lattice capacity), not as $g'^2/(g^2+g'^2)$. This is a *departure from the SM definition*, justified by the lattice framework where mode counts are topological invariants and efficiencies are metric-dependent. The denominator is fixed by topology; only the numerator carries the metric correction.

6. **K_3 eigenvalue ratio gives** $K = 2/3$ (Eq. 12). This is *proved* in SM-3.

7. **Self-energy isotropy on K_3 faces** (Theorem 4.2). This is *proved* by explicit computation.

8. **Bond counting for** ε (Steps A–D). Each step uses 600-cell geometry (2 internal bonds, $z+1 = 13$) and the proved isotropy theorem. This is *derived*.

9. **Closed-neighbourhood normalisation** $z+1 = 13$ (Step C). The normalisation uses the closed-neighbourhood Laplacian $\tilde{L} = (z+1)I - \tilde{A}$. Numerically, $z+1 = 13$ matches PDG to 0.003%; $z = 12$ matches to 0.021% (7× worse). This is *standard in lattice field theory* and *numerically confirmed*.

The weakest link is assumption 5: the operational definition of the Weinberg angle in CPP. This is a genuine departure from the SM and is the point where a skeptical reader would push hardest. The justification is that CPP is a lattice theory where couplings are emergent, not fundamental, and the natural mixing variable is the mode fraction, not the coupling ratio.

8 Conclusion

The entire charged lepton mass spectrum follows from the 600-cell lattice geometry:

$$\begin{gathered}\textbf{6 axioms} \to \mathrm{K}_3 \text{ eigenvalue ratio} \to K = 2/3 \\ \to \text{spectral traces} \to \text{bare fraction } 3/8 \\ \to \text{edge efficiency } 1/\varphi \to \sin^2\theta_W = 3/(8\varphi) \\ \to \text{bond counting} + \text{isotropy} \to \varepsilon = 3/(52\varphi) \\ \to \cos\theta = -(2/3)(1 + 3/(104\varphi)) \\ \to \text{lepton masses (1 calibration, 0 shape parameters)}\end{gathered}$$

The Koide ratio and the Koide phase are *the same number*: $K = 2/3$ appears once as a fraction and once as a cosine, both from the K_3 eigenvalue ratio $\lambda_+/|\lambda_-| = 2$. The Weinberg angle is the geometric mixing ratio of abelian (edge) and non-abelian (face) propagation modes on the vacuum lattice.

The Standard Model requires three free parameters for the charged lepton masses. This derivation requires one.

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